THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS TWO-CHAPTER TEST SET D

Chapters 11 & 12: **Exploring Some Geometric Themes, Tales by Dots and Lines** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The Koch snowflake is built by repeatedly adding triangles to: 1

(a) the interior **(b)** the vertices
(c) the middle third of each side **(d)** the base only

2. Which solid has a circular face and one vertex? 1

(a) cone **(b)** sphere **(c)** cylinder **(d)** cuboid

3. If a cube is cut by a plane parallel to a face, the cross section is a: 1

(a) hexagon **(b)** square **(c)** triangle **(d)** circle

4. The Sierpinski triangle is obtained by repeatedly removing: 1

(a) one corner **(b)** the base
(c) the central inverted triangle **(d)** a vertex

5. A solid with two parallel congruent polygonal bases joined by rectangles is a: 1

(a) pyramid **(b)** cone **(c)** sphere **(d)** prism

6. If every value in a data set is increased by 5, the mean: 1

(a) increases by 5 **(b)** increases by 25 **(c)** stays the same **(d)** doubles

7. In a line graph, a steep upward line segment shows: 1

(a) a decrease **(b)** no change **(c)** an error **(d)** a rapid increase

8. If the mean of 10 numbers is 15, their sum is: 1

(a) 150 **(b)** 1.5 **(c)** 25 **(d)** 15

9. The mean of 2, 4, 6, 8, 10 is: 1

(a) 30 **(b)** 6 **(c)** 8 **(d)** 5

10. The range of 2, 3, 3, 5, 3, 7 is: 1

(a) 7 **(b)** 4 **(c)** 5 **(d)** 3

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. Which solid has no vertex and no edge? 2

12. Name the front view of a cube. 2

13. A line graph shows temperature rising from 20°C to 30°C between 6 a.m. and 12 noon. Find the average rise per hour. 2

14. If the mean of $x, x+2, x+4$ is 10, find x. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Why can a regular pyramid with base 7 sides have 8 faces? 3

16. The median of 5 numbers 4, 7, x, 12, 15 is 9. Find x and the mean. 3

17. Match each solid with its top view: cube, cylinder, cone, square pyramid. 3

OR

State two uses of nets in everyday life. 3

18. Show that the mean of a and b lies between them. 3

OR

Two students' scores in 4 tests are A: 60, 65, 70, 75 and B: 50, 60, 75, 85. Compare means and ranges. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. (i) How do fractals appear in nature and art?
(ii) Give two examples. 5

OR

Case study: An architect draws a plan of a pyramid-shaped monument. 5
(i) Name its faces.
(ii) The base is a square of side 40 m, height 30 m; find the slant height.
(iii) Find the area of one triangular face.

20. **Case study:** A cricketer's runs in 8 matches: 12, 45, 0, 38, 62, 18, 45, 90. 5
(i) Find mean and median.
(ii) Find range.
(iii) Comment.

OR

(i) Four numbers have mean 12. If three of them are 8, 10 and 14, find the fourth. 5
(ii) Find the median of the four numbers.

END OF QUESTION PAPER — Total: 40 Marks

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